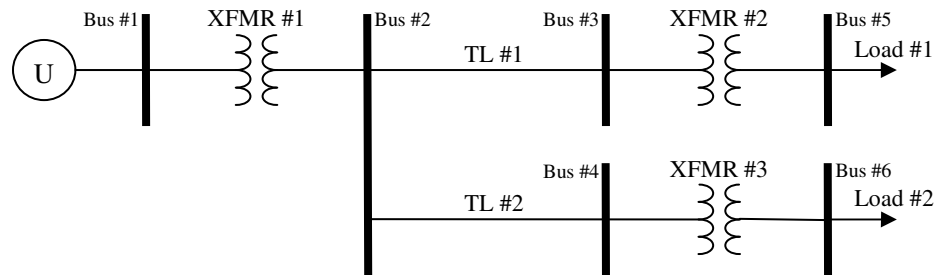


Problem #1 (10-points)

The one line diagram for a single phase power system is drawn below. Using the series impedance model for TL#1 and the pi model for TL#2, **draw an equivalent circuit for this power system.** (use either portrait mode or landscape mode)



Problem #2 (25-points)

Steps 1, 2, and 3 of the PSA procedure were done for a three phase power system, and are shown below. **Perform steps 4 and 5 of the PSA procedure to calculate the generator real and reactive power output.**

Step 1: Visualize	Step 2: Equivalent Circuit
Step 3: Circuit Parameters $\bar{E}_G = \left(\frac{243.85}{\sqrt{3}} \right) \angle 0^\circ \text{ V}, \quad \bar{Z}_G = 0.02 \angle 86^\circ \Omega, \quad \bar{Z}_{L1} = \bar{Z}_{YL1} = 5 \angle 75^\circ \Omega, \quad \bar{Z}_{L2} = \frac{1}{3} \bar{Z}_{\Delta L2} = 3 \angle -25^\circ \Omega$ $\bar{Z}_{TL} = (450 \text{ ft}) (0.0004 \angle 57^\circ \Omega / \text{ft}) = 0.18 \angle 57^\circ \Omega$	

Problem #3 (20-points)

In a small single phase power system, four 300V distribution panels are located at different locations in a large building. The first distribution panel connects to the second distribution panel through a 250ft power cable. The first distribution panel also connects to the fourth distribution panel through a 500ft power cable. The third distribution panel connects to the second distribution panel through a 300ft power cable. The third distribution panel also connects to the fourth distribution panel through a 150ft power cable. At each distribution panel, there is a step down transformer which converts the nominal 300V level to a nominal 120V level required by the system loads. A load is connected to the 120V side of each of these transformers. At the first 300V distribution panel, there is another step-down transformer used to connect a 480V electric utility tie to this 300V panel. The system is normally powered from the utility tie, but there is also a 300V emergency backup generator installed at the fourth distribution panel. This generator will power the system if the utility tie is lost. **Draw a block diagram for this power system.**

Problem #4 (30-points)

In a single phase power system, an ideal source supplies a load through a 400ft length of 2-conductor power cable. The source operates at 487V and supplies 1212W, 2080VAR. The power cable has a per conductor distributed series impedance of $0.0013 \angle 25^\circ \Omega/\text{ft}$ and a distributed shunt admittance of $0.00001 \angle 87^\circ \text{ S}/\text{foot}$ between the two conductors. **Using 5-step PSA, calculate the power cable efficiency.** Use the pi model for the power cable.

Problem #5 (8-points)

A three phase Y-connected generator rated 500KVA, 480V had an equivalent circuit impedance of $0.125 \angle 85^\circ \Omega$ calculated for it in PSA step 3. Calculate the **stator winding reactance in %** and **X/R ratio** that would appear in the specifications for this generator.

Problem #6 (7-points)

A three phase Δ -Y connected transformer rated 750KVA, 13.8KV/480V had an equivalent circuit impedance of $10.261 + j28.191 \Omega$ calculated for it in PSA step 3. Calculate the **impedance in %** and **X/R ratio** that would appear in the specification for this transformer.